

Chaehyeong Lee

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Research Interests

Geofluid Dynamics: cross-scale upper-ocean mixing processes; Mesoscale eddies.

Physical Oceanography: long-term change in the upper-ocean thermal state; ocean's role in climate systems.

Education

University of Colorado Boulder

Boulder, CO, USA

Ph.D. in Atmospheric and Oceanic Sciences

Aug. 2024 – present

Advisor: Dr. Donata Giglio & Co-Advisor: Dr. Aneesh Subramanian

Yonsei University

Seoul, Rep. of Korea

M.S. in Atmospheric Sciences

Aug. 2023

Advisor: Dr. Hajoon Song

Yonsei University

Seoul, Rep. of Korea

B.S. in Atmospheric Sciences

Feb. 2022

Advisor: Dr. Hajoon Song

Publications

Published.....

Lee, C., Song, H., Choi, Y., Cho, A., & Marshall, J. (2025). Observed Multi-decadal Increase in the Surface Ocean's Thermal Inertia. *Nature Climate Change*, **15**, 308-314. doi:10.1038/s41558-025-02245-w

Under Review.....

Lee, C., Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., Molod, A., & Ruiz-Xomochuk, V. (2026). Effects of Satellite Sea Surface Salinity Assimilation on the Representation of Upper-Ocean Thermohaline Structure and Variability in the NASA GEOS-S2S-2 model. *Quarterly Journal of the Royal Meteorological Society*

In preparation.....

Lee, C., Giglio, D., Subramanian, A. C., Mazloff, M., & Arbic, B. K. Ocean Advection Drives Mixed Layer Temperature Variability in Energetic Regions of the Global Ocean.

Research Experience

Dr. Giglio's Research Group, CU Boulder

Boulder, CO, USA

Research Assistant

Aug. 2024 – present

Climate Processes and Predictability Group, CU Boulder

Boulder, CO, USA

Research Assistant

Aug. 2024 – present

Teaching Experience

Data Science Laboratory, taught by Prof. Donata Giglio at CU Boulder <i>Teaching Assistant</i>	Boulder, CO, USA <i>Spring 2026</i>
Climate & Civilization, taught by Prof. Yign Noh at Yonsei Univ. <i>Teaching Assistant</i>	Seoul, Rep. of Korea <i>Spring 2023</i>
Physical Oceanography, taught by Prof. Hajoon Song at Yonsei Univ. <i>Teaching Assistant</i>	Seoul, Rep. of Korea <i>Fall 2022</i>

Awards & Scholarships

2025: ATOC Fellowship, Department of Atmospheric and Oceanic Sciences, University of Colorado Boulder
2025: Academic Research Grants (GCP research credits), Google LLC
2024: Outstanding Thesis Award, Yonsei University Graduate School, Yonsei University
2022–2023: Full tuition merit scholarship (for the top 2 graduate students), Yonsei University
2022: High Honors for academic performance, Yonsei University
2020–2021: Jilli Scholarship for academic performance, Yonsei University

Patent

Song, H., & Lee, C. (2025). *Evaluation system and method of persistence of SST anomalies using autocorrelation coefficient and arctangent regressive model*. Rep. of Korea Patent #1028135790000. doi:10.8080/1020220157159

Invited Talk

NASA Salinity Telecon <i>Assessing the Impact of Satellite Sea Surface Salinity Assimilation on Vertical Structure of the Upper Ocean in the NASA GEOS-S2S 2.</i>	Virtual Meeting <i>Dec. 2025</i>
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Selected Presentations

2027 American Meteorological Society Annual Meeting (poster) <i>Ocean Advection Drives Mixed Layer Temperature Variance at 10-100 km in the Gulf Stream Extension.</i> Lee, C., Giglio, D., & Subramanian, A. C.	Denver, CO, USA <i>Jan. 2027 (scheduled)</i>
US Climate Variability and Predictability (poster) <i>Enhanced Upper-Ocean Horizontal Advection in a High-Resolution Climate Model Improves the Representation of High-Frequency Near-Surface Temperature Variability.</i> Lee, C., Giglio, D., & Subramanian, A. C.	Boulder, CO, USA <i>Sep. 2026 (scheduled)</i>
Ocean Science Meeting 2026 (poster) <i>Assessing the Impact of Satellite Sea Surface Salinity Assimilation on the Upper Ocean Thermal State in the NASA GEOS S2S-v2 Model.</i> Lee, C., Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., & Molod, A.	Glasgow, Scotland <i>Feb. 2026</i>

American Geophysical Union Fall Meeting (poster) <i>The Increasing Trend of Persistence of Sea Surface Temperature in the Past 40 Years.</i> Lee, C., Song, H., Cho, A., & Tak, Y.	Chicago, IL, USA Dec. 2022
Korean Society of Oceanography Spring Conference (talk) <i>Increasing Persistence of SST Anomalies and Duration of Marine Heatwaves.</i> Lee, C., Song, H., Cho, A., & Tak, Y.	Jeju, Rep. of Korea Jun. 2022

Workshops

Broadening Earth System Research Across Scales with High-Resolution Modeling Workshop <i>US Climate Variability and Predictability</i>	Boulder, CO, USA Sep. 2026 (scheduled)
Data Assimilation Training and Science Workshop <i>Consortium for Advanced Data Assimilation Research and Education & NOAA EPIC</i>	Norman, OK, USA Jun. 2026
14th International Workshop on Modeling the Ocean 2024 <i>International Workshop on Modeling the Ocean</i>	Sapporo, Japan Jun. 2024
User Training for the Glosea 6 Climate Prediction Model <i>Korea Meteorological Administration</i>	Jeju, Rep. of Korea Jan. 2022
Deep Learning Training: Fundamentals of Deep Learning <i>NVIDIA Deep Learning Institute</i>	Gonju, Rep. of Korea Jan. 2022

Service

Peer Reviewer: *Journal of Climate*

Technical Skills

Programming: Python (xarray, dask, Pangeo), Julia
Modeling: MITgcm, Oceananigans
HPC: Parallel/distributed computing, NCAR Casper/Derecho clusters
Tools: Git, Linux shell scripting, LaTeX